

[Secure Exit](#)SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

CHITTIBOINA HARSHITHA
192473031RA

CO3-AT1-Assignment

[View Only](#)[← Back](#)

QUESTIONS

Q1Banking Transaction System –
Exception Handling
10 marks**Q2**Hospital Patient Registration –
User-Defined Exception
10 marks**Q3**E-Commerce Order Processing
– Built-in Exceptions
10 marks**Q4**Online Food Delivery – Multiple
Threads and Execution Order
10 marks**Q5**Manufacturing Plant –
Producer-Consumer Problem
10 marks

5/5 answered

Q3

E-Commerce Order Processing – Built-in Exceptions

10
marks

Develop a Java program for an online shopping application. Store product names and prices in arrays and accept a product index and quantity from the customer.

The program must:

- Calculate total cost as quantity × price.
- Handle invalid product indices.
- Handle invalid numeric input.
- Reject zero or negative quantities.
- Use appropriate built-in exception classes.
- Continue execution after handling an invalid transaction.

Sample Test Cases:

Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case

5, 2 ArrayIndexOutOfBoundsException handled

Boundary / exception case

2, -1 Invalid quantity Invalid input

abc, 2 InputMismatchException handled Invalid input

Your Code

```
1 import java.util.Scanner;
2 public class ecommerceorderprocessing
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         String[] pro = {"mobile", "laptop", "t"};
8         double[] prices = {20000, 70000, 50000};
9         try
10        {
11            int index = sc.nextInt();
12            int quantity = sc.nextInt();
13            if(quantity <= 0)
14            {
15                throw new IllegalArgumentException();
16            }
17            System.out.println(pro[index]);
18            System.out.println(prices[index]);
19        }
20        catch(ArrayIndexOutOfBoundsException)
21        {
22            System.out.println("arrayindexoutofbounds");
23        }
24        catch(IllegalArgumentException e)
25        {
26            System.out.println("invalid quantity");
27        }
28        catch(Exception e)
29        {
30            System.out.println("invalid input");
31        }
32    }
33 }
```

Input

2
2

Output

```
tv  
100000.0
```

✓ Submitted

© 2026 **SIMATS Online Lab**. All rights reserved.
